

	ELEC 205 Assignment 4 ASM Design Problem
	Deadline: May 29, 11:59 PM
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Task 1

"+": a full adder with 4-bit and returns 4-bit result and final carry

x: $x_3x_2x_1x_0$ (input)

y: $y_3y_2y_1y_0$ (input)

result: $r_3r_2r_1r_0$

regx, regy: output of registers

o: o_1o_0

l: load

function main:

if o==00 and l==1:

regx, regy $\leftarrow$ x, y

result, carry $\leftarrow$ regx + regy

overflow $\leftarrow$ overflowDetection(regx, regy, result)

if o==01 and l==1:

regx, regy $\leftarrow$ x, y

result, carry $\leftarrow$ regx + regy' + 1

overflow $\leftarrow$ overflowDetection(regx, regy, result)

if o==10 and l==1:

regx, regy $\leftarrow$ x, y

regx $\leftarrow$ sl regx

regy $\leftarrow$ sr regy

result, carry $\leftarrow$ regx + regy

overflow $\leftarrow$ overflowDetection(regx, regy, result)

if o==11 and l==1:

regx, regy $\leftarrow$ x, y

regx $\leftarrow$ sr regx

regy $\leftarrow$ sr regy

result, carry $\leftarrow$ regx + regy' + 1

overflow $\leftarrow$ overflowDetection(regx, regy, result)

function overflowDetection(regx, regy, result):

return (MSB(regx)==MSB(regy) but != MSB(result))

Task 2

Task 3

Task 4

Task 5

Task 6